



KA999

Alumina (Al_2O_3) 99.9%

KEY PROPERTIES

- Excellent abrasion resistance
- Excellent dielectric properties
- Improved corrosion resistance
- High thermal conductivity
- Improved thermal shock resistance

APPLICATIONS

- Labware
- Mineral grinding wear
- Machinery parts
- Chemical industry
- High voltage dielectric components

MAIN PROPERTIES

Property	Value	Unit
Density	3.9	g/cm^3
Flexural Strength	400	MPa
Hardness	18	GPa
Fracture Toughness	3.5	$\text{MPa}\cdot\text{Vm}$
Young Modulus	370	GPa
Thermal Conductivity (100°C)	35	$\text{W/m}\cdot\text{K}$
CTE (20–1000°C)	8.2	$\times 10^{-6} \text{ K}^{-1}$

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)	1600	°C
Recomended temp	<1550	°C
Continuous temp	1500	°C

ELECTRICAL BEHAVIOR

Resistivity (25°C)	>1E14	$\Omega\cdot\text{cm}$
Dielectric strength	16	kV/mm

MATERIAL INFORMATION

Material type	Al_2O_3
Machinability	Low
Structure	Dense ceramic
Purity	99.9%
Color	White / ivory

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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